

MAHEK PATEL

Software Engineer · mppatel12@gmail.com · github.com/mpate1001 · linkedin.com/in/mahek-patel

Software engineer with 7+ years building data platforms at scale. Currently building real-time data pipelines on Kafka, KSQL, and AWS at Duke Energy. Background spans streaming infrastructure, ETL/ELT, SQL engineering, and applied machine learning.

EXPERIENCE

IT Software Engineer — Duke Energy Inc. June 2025 – Present

- Build data pipelines extracting Oracle Maximo data and streaming it through Kafka connectors and KSQL for real-time consumption
- Design and maintain Kafka Connect configurations for hundreds of tables, enabling customers to consume enterprise data
- Deploy and manage data infrastructure on AWS, delivering processed data to PostgreSQL databases
- Author and maintain Terraform configurations for all AWS infrastructure provisioning and management
- Monitor consumer lag, throughput, and p99 latency; handle scaling and rolling upgrades

Tech: Kafka · KSQL · Kafka Connect · Terraform · Oracle · PostgreSQL · AWS

Senior IT Business Application Analyst — Duke Energy Inc. Jan 2020 – June 2025

- Supported CATSWeb compliance application, translating business requirements into data models and exports
- Engineered T-SQL stored procedures, views, and export jobs in SSMS with robust validation
- Built PowerShell + Selenium automation for user provisioning and admin tasks
- Developed Power BI dashboards surfacing compliance KPIs and anomaly flags
- Managed incident triage in Jira/ServiceNow, stabilizing nightly loads and reporting SLAs

Tech: T-SQL · SSMS · PowerShell · Selenium · Power BI · Jira

Business Analyst — Lowe's Companies Inc. 2018 – 2020

- Supervised incentivized promotions for GE Lighting products offered through government programs via Utility companies
- Arranged, formatted, and uploaded pricing data into the Lowe's database using MS Access
- Generated transactional reports to inform decisions on high-priority incentive discount programs across all locations
- Prepared reports for Accounting team to ensure accurate invoicing to GE Lighting
- Extracted reports using Access commands to analyze current and historical pricing data

Tech: MS Access · Excel · SQL · Data Analysis

Business Analyst — Spectrum Reach 2017 – 2018

- Directed the rate card portfolio process, organizing, formatting, and publishing rate cards for the sales team
- Worked with field sales and Pricing & Planning Manager to update and adjust rates on regular and ad hoc basis
- Coordinated rate cards for multiple channels including local, national, and political
- Leveraged MySQL to extract past data for comprehensive pricing analysis and transferred data to Excel

Tech: MySQL · Excel · Data Analysis

Business Owner - Real Estate — Akshar Raj LLC

2017 – Present

- Own and manage residential & Commercial real estate properties generating steady monthly rental income
- Handle property acquisition, tenant relations, and ongoing maintenance coordination
- Manage financial operations including budgeting, expense tracking, and tax planning

Tech: Property Management · Financial Planning

EDUCATION

UNC Chapel Hill

2024 – May 2026

Master of Science in Data Science — Machine Learning · Statistical Analysis · Data Engineering

UNC Charlotte

2019 – 2021

Master of Science in Data Science and Business Analytics — GPA: 3.9 · Business Analytics · Data Mining

UNC Charlotte

2013 – 2017

Bachelor of Science in Business Administration & MIS — Management Information Systems · Business Administration

SKILLS

Languages

Python · SQL · JavaScript · Java · KSQL · PowerShell

Data & ML

ETL/ELT · Apache Kafka · Stream Processing · Avro/Schema Registry · Pandas · Power BI

Engineering

Microservices · REST APIs · AWS (RDS, S3) · Terraform · CI/CD · GitHub Actions

Tools &

Databases

PostgreSQL · MySQL · SQL Server · Git · Jira · Selenium